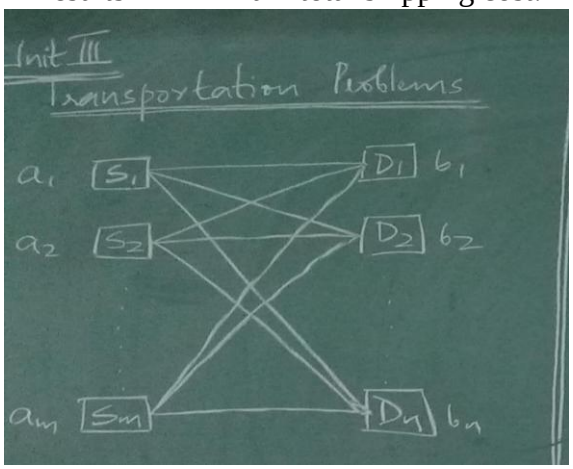
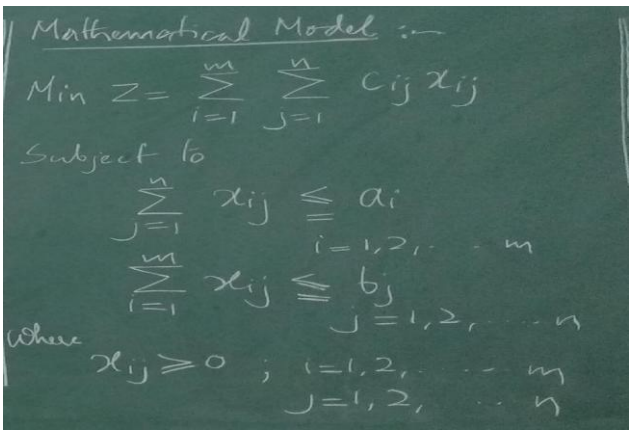
Unit – III

TRANSPORTATION PROBLEMS

The transportation problem is a special type of linear programming problem, where the objective is to minimize the cost of distributing a product from a number of sources to a number of destinations.

The transportation problem deals with a special class of linear programming problems in which the objective is to transport a homogeneous product manufactured at several plants (origins) to a number of different destinations at a minimum total cost. The total supply available at the origin and the total quantity demanded by the destinations are given in the statement of the problem. The cost of shipping a unit of goods from a known origin to a known destination is also given. Our objective is to determine the optimal allocation that results in minimum total shipping cost.





General Mathematical model:

Minimize $Z = \sum c_{ij}x_{ij}$ where i = 1,2, ..., m and j = 1,2, ..., n

Subject to

$\sum x_{ij} \leq Si$ for j = 1,2, ..., n & i = 1,2, ..., m (supply)

$\sum x_{ij} \geq Dj$ for i = 1,2, ..., m & j = 1,2, ..., n (demand)

$x_{ij} \geq 0$

For a feasible solution to exist, it is necessary that total capacity equals total requirements.

Total supply = total demand.

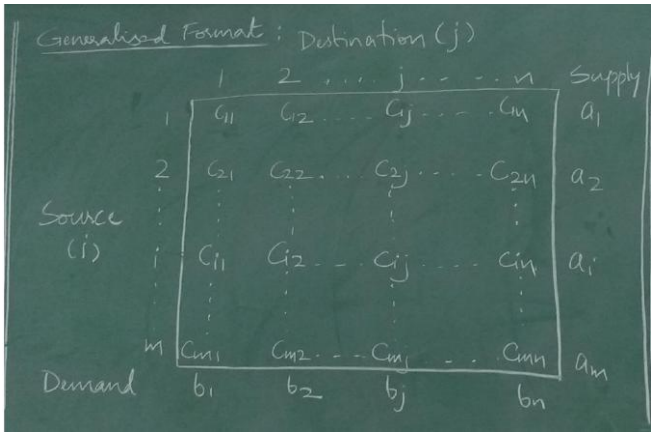
or

$\sum a_i = \sum b_j$

Basic Terminology:

1. ***Source(Origin):*** It is the location from which shipments are dispatched.
2. ***Destination:*** It is the location to which shipments are transported.
3. ***Unit Transportation cost:*** It is the cost of transporting one unit of the consignment from an origin to a destination.
4. ***Perturbation Technique:*** It is a method used for modifying a degenerate transportation problem, so that the degeneracy can be resolved.
5. ***Feasible Solution:*** A solution that satisfies the row and column sum restrictions and also the non-negativity restrictions is a feasible solution.
6. ***Basic Feasible Solution:*** A feasible solution of (m x n) transportation problem is said to be basic feasible solution, when the total number of allocations is equal to (m + n – 1).

7. ***Optimal Solution:*** A feasible solution is said to be optimal solution when the total transportation cost will be the minimum cost.



Types of Transportation Problem:

i. Balanced Transportation Problem:

$\sum a_i = \sum b_j$

ii. Unbalanced Transportation Problem

$\sum a_i$ Not equal to $\sum b_j$

		Destination			Supply
		1	2	3	
Source	1	30	50	15	300
	2	35	70	20	200
	3	20	45	60	500
Demand		300	200	400	

Supply = 1000 and Demand = 900

		Destination				Supply
		1	2	3	Dummy Column 4	
Source	1	30	50	15	0	300
	2	35	70	20	0	200
	3	20	45	60	0	500
Demand		300	200	400	100	

Methods for finding an Initial Basic Feasible Solution (IBFS):

i. North West Corner Rule
ii. Least Cost Entry Method
iii. Row Minima Method
iv. Column Minima Method
v. Vogel's Approximation Method

i. **North West Corner Rule:** The North West corner rule is a method for computing a basic feasible solution of a transportation problem, where the basic variables are selected from the North – West corner (i.e., top left corner).

Steps:

a. Select the north west (upper left-hand) corner cell of the transportation table and allocate as many units as possible equal to the minimum between available supply and demand, i.e., min(s_1, d_1).
b. Adjust the supply and demand numbers in the respective rows and columns.
c. If the demand for the first cell is satisfied, then move horizontally to the next cell in the second column.
d. If the supply for the first row is exhausted, then move down to the first cell in the second row.
e. If for any cell, supply equals demand, then the next allocation can be made in cell either in the next row or column.
f. Continue the process until all supply and demand values are exhausted.

ii. **Least Cost Entry Method:** Least cost entry method is a method for computing a basic feasible solution of a transportation problem, where the basic variables are chosen according to the unit cost of transportation. This method is very useful because it reduces the computation and the time required to determine the optimal solution. The following steps summarize the approach.

Steps:

a. Identify the box having minimum unit transportation cost (c_{ij}).
b. If the minimum cost is not unique, then you are at liberty to choose any cell.
c. Choose the value of the corresponding x ij as much as possible subject to the capacity and requirement constraints.
d. Repeat steps a - c until all restrictions are satisfied.

iii. **Vogel's Approximation Method:** The Vogel approximation (Unit penalty) method is an iterative procedure for computing a basic feasible solution of a transportation problem. This method is preferred over the two methods discussed in the previous sections, because the initial basic feasible solution obtained by this method is either optimal or very close to the optimal solution.

Steps:

a. Identify the boxes having minimum and next to minimum transportation cost in each row and write the difference (penalty) along the side of the table against the corresponding row.
b. Identify the boxes having minimum and next to minimum transportation cost in each column and write the difference (penalty) against the corresponding column.
c. Identify the maximum penalty. If it is along the side of the table, make maximum allotment to the box having minimum cost of transportation in that row. If it is below the table, make maximum allotment to the box having minimum cost of transportation in that column.
d. If the penalties corresponding to two or more rows or columns are equal, you are at liberty to break the tie arbitrarily.
e. Repeat the above steps until all restrictions are satisfied.

Problem: Find the Initial Basic feasible Solution by all methods for the following transportation problem

		Warehouse				Factory Capacity
		W_1	W_2	W_3	W_4	
Factory	F_1	19	30	50	10	7
	F_2	70	30	40	60	9
	F_3	40	8	70	20	18
Warehouse Requirement		5	8	7	14	

NWCR

		Warehouse				Factory Capacity
		W_1	W_2	W_3	W_4	
Factory	F_1	19(5)	30(2)	50	10	~~7~~ ~~2~~ 0
	F_2	70	30(6)	40(3)	60	~~9~~ ~~3~~ 0
	F_3	40	8	70(4)	20(14)	~~18~~ ~~14~~ 0
Warehouse Requirement		~~5~~ 0	~~8~~ ~~6~~ 0	~~7~~ ~~4~~ 0	~~14~~ 0	34

Total Transportation Cost = Rs. (19x5+30x2+30x6+40x3+70x4+20x14)
= Rs. 1015/-

RMM

		Warehouse				Factory Capacity
		W_1	W_2	W_3	W_4	
Factory	F_1	19	30	50	10(7)	~~7~~ 0
	F_2	70	30(8)	40(1)	60	~~9~~ ~~1~~ 0
	F_3	40(5)	8	70(6)	20(7)	~~18~~ ~~11~~ ~~6~~ 0
Warehouse Requirement		~~5~~ 0	~~8~~ 0	~~7~~ ~~6~~ 0	~~14~~ ~~7~~ 0	34

Total Transportation Cost = Rs. (10x7+30x8+40x1+40x5+70x6+20x7)
= Rs. 1110/-

CMM

		Warehouse				Factory Capacity
		W_1	W_2	W_3	W_4	
Factory	F_1	19(5)	30	50	10(2)	~~7~~ ~~2~~ 0
	F_2	70	30	40(7)	60(2)	~~9~~ ~~2~~ 0
	F_3	40	8(8)	70	20(10)	~~18~~ ~~10~~ 0
Warehouse Requirement		~~5~~ 0	~~8~~ 0	~~7~~ 0	~~14~~ ~~12~~ ~~2~~ 0	34

Total Transportation Cost = Rs. (19x5+10x2+40x7+60x2+8x8+20x10)
= Rs. 779/-

LCEM

		Warehouse				Factory Capacity
		W_1	W_2	W_3	W_4	
Factory	F_1	19	30	50	10(7)	~~7~~ 0
	F_2	70(2)	30	40(7)	60	~~9~~ ~~2~~ 0
	F_3	40(3)	8(8)	70	20(7)	~~18~~ ~~10~~ ~~3~~ 0
Warehouse Requirement		~~5~~ ~~2~~ 0	~~8~~ 0	~~7~~ 0	~~14~~ ~~7~~ 0	34

Total Transportation Cost = Rs. (10x7+70x2+40x7+40x3+8x8+20x7)
= Rs. 814/-

Vogel's Approximation Method (VAM)

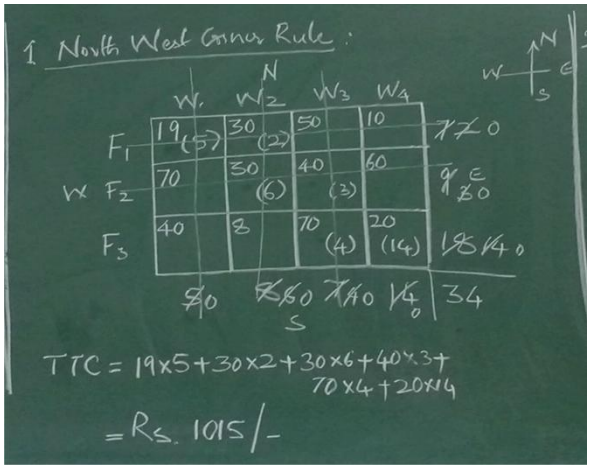
		Warehouse				Factory Capacity	Row Penalties				
		W_1	W_2	W_3	W_4						
Factory	F_1	19(5)	30	50	10(2)	~~7~~ ~~2~~ 0	9	9	40	40	
	F_2	70	30	40(7)	60(2)	~~9~~ ~~7~~ 0	10	20	20	20	20
	F_3	40	8(8)	70	20(10)	~~18~~ ~~10~~ 0	12	20	50	←	
Warehouse Requirement		~~5~~ 0	~~8~~ 0	~~7~~ 0	~~14~~ ~~4~~ ~~2~~ 0	34					
Column Penalties		21	22	10	10						
		21		10	10						
				10	10						
				10	50						
				40	60						

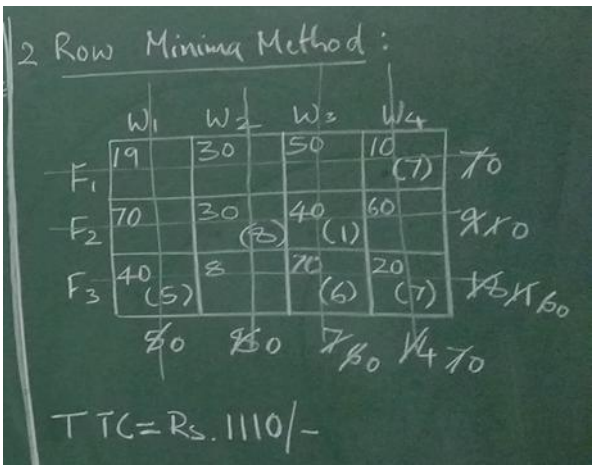
Total Transportation Cost = Rs. (19x5+10x2+40x7+60x2+8x8+20x10)
= Rs. 779/-

Find the IBFS by all the methods for the following transportation problem:

		Destination				Supply
		I	II	III	IV	
Source	A	21	16	15	13	11
	B	17	18	14	13	13
	C	32	27	18	41	19
Demand		6	10	12	15	

NWCR - Rs. 1095/- LCEM - Rs. 837
RMM - Rs. 837/- VAM - 646/-
CMM - Rs. 1062/-





Problem: Find the Initial Basic feasible Solution of the following transportation problem by Vogel's Approximation Method:

		Destination				Supply
		1	2	3	4	
Origin	1	20	22	17	4	120
	2	24	37	9	7	70
	3	32	37	20	15	50
Demand		60	40	30	110	

Solution: Calculating penalty for table 1
17 - 4 = 13, 9 - 7 = 2, 20 - 15 = 5
24 - 20 = 4, 37 - 22 = 15, 17 - 9 = 8, 7 - 4 = 3

Table 1

		Destination				Supply	Row Penalty
		1	2	3	4		
Origin	1	20	22 (40)	17	4	~~120~~ 80	13
	2	24	37	9	7	70	2
	3	32	37	20	15	50	5
Demand		60	40	30	110		
Column Penalty		4	15	8	3		

The highest penalty occurs in the second column. The minimum c_{ij} in this column is c_{12} (i.e., 22). So x_{12} = 40 and the second column is eliminated. The new reduced matrix is shown below and again calculate the penalty:

Table 2

		1	3	4	Supply	Row Penalty
Origin	1	20	17	4 (80)	~~120~~ 40	13
	2	24	9	7	70	2
	3	32	20	15	50	5
Demand		60	30	110		
Column Penalty		4	8	3		

The highest penalty occurs in the first row. The minimum c_{ij} in this row is c_{14} (i.e., 4). So x_{14} = 80 and the first row is eliminated. The new reduced matrix is shown below:

Table 3

Origin	1	3	4	Supply	Penalty
2	24	9 (30)	7	70	2
3	32	20	15	50	5
Demand	60	~~30~~	30		
Penalty	8	11	8		

The highest penalty occurs in the second column. The minimum cij in this column is c_{23} (i.e., 9). So x_{23} = 30 and the second column is eliminated. The reduced matrix is given in the following table.

Table 4

Origin	1	4	Supply	Penalty
2	24 (10)	7 (30)	~~40~~	17
3	32 (50)	15	~~50~~	17
Demand	~~60~~	~~30~~		
Penalty	8	8		

The following table shows the computation of penalty for various rows and columns.

		Destination				Supply	Row Penalty					
		1	2	3	4							
Origin	1	20	22 (40)	17	4 (80)	~~120~~	13	13	-	-	-	-
	2	24 (10)	37	9 (30)	7 (30)	~~70~~	2	2	2	17	24	24
	3	32 (50)	37	20	15	~~50~~	5	5	5	17	32	-
Demand		~~60~~	~~40~~	~~30~~	~~110~~	240						
Column Penalty		4	15	8	3							
		4	-	8	3							
		8	-	11	8							
		8	-	-	8							
		8	-	-	-							
		24	-	-	-							

Modified Distribution Method (MODI) or (u - v) Method:

The modified distribution method, also known as MODI method or (u - v) method provides a minimum cost solution to the transportation problem. In the stepping stone method, we have to draw as many closed paths as equal to the unoccupied cells for their evaluation. To the contrary, in MODI method, only closed path for the unoccupied cell with highest opportunity cost is drawn.

Steps:

i. Determine an initial basic feasible solution using any one of the three methods given below:
   a. North West Corner Rule
   b. Matrix Minimum Method
   c. Vogel Approximation Method

ii. Determine the values of dual variables, u_i and v_j , using $u_i + v_j = c_{ij}$

ii. Compute the opportunity cost using $c_{ij} – (u_i + v_j)$.

iii. Check the sign of each opportunity cost. If the opportunity costs of all the unoccupied cells are either positive or zero, the given solution is the optimal solution. On the other hand, if one or more unoccupied cell has negative opportunity cost, the given solution is not an optimal solution and further savings in transportation cost are possible.

iv. Select the unoccupied cell with the smallest negative opportunity cost as the cell to be included in the next solution.

v. Draw a closed path or loop for the unoccupied cell selected in the previous step. Please note that the right angle turn in this path is permitted only at occupied cells and at the original unoccupied cell.

vi. Assign alternate plus and minus signs at the unoccupied cells on the corner points of the closed path with a plus sign at the cell being evaluated.

vii. Determine the maximum number of units that should be shipped to this unoccupied cell. The smallest value with a negative position on the closed path indicates the number of units that can be shipped to the entering cell. Now, add this quantity to all the cells on the corner points of the closed path marked with plus signs, and subtract it from those cells marked with minus signs. In this way, an unoccupied cell becomes an occupied cell.

viii. Repeat the whole procedure until an optimal solution is obtained.

Problem: Obtain an optimal solution to the transportation problem by MODI method given in table below:

		To				Supply
		1	2	3	4	
From	1	2	3	11	7	6
	2	1	0	6	1	1
	3	5	8	15	9	10
Demand		7	5	3	2	

Solution:

Step I: Make a transportation model

		To				Supply
		1	2	3	4	
From	1	2	3	11	7	6
	2	1	0	6	1	1
	3	5	8	15	9	10
Demand		7	5	3	2	

Step II: Find the initial basic feasible solution (If not mentioned use VAM)

		To				Supply
		1	2	3	4	
From	1	2(1)	3(5)	11	7	6
	2	1	0	6	1(1)	1
	3	5(6)	8	15(3)	9(1)	10
Demand		7	5	3	2	

The total transportation cost = Rs. 102/-

Step III: Perform Optimality Test

It can be performed only on that feasible solution in which

a. m+n-1 = no. of allocations

3+4-1 = 6

6 = 6 √

where m - no. of rows & n - no. of column

b. These m+n-1 allocations should be in independent

In this the allocated values does not form any loop.

Modified Distribution Method (MODI) or (u - v) Method:

Step i: Set up a cost matrix containing the costs associated with the cells for which allocations have been made.

2	3		
			1
5		15	9

Step ii: Enter a set of number v_j across the top of the matrix & a set of number u_i across the left side so that their sums equal the costs entered in step i.

	v_1=0	**v_2=1**	**v_3=10**	**v_4=4**
u_1=2	2	3		
u_2=-3				1
u_3=5	5		15	9

Step iii: Fill the vacant cells with the sums of u_i & v_j.

	v_1=0	v_2=1	v_3=10	v_4=4
u_1=2	.	.	12	6
u_2=-3	-3	-2	7	.
u_3=5	.	6	.	.

Step iv: Subtract the cell values of the matrix of step iii from the original cost matrix. The resulting matrix is called **C**ell **E**valuation **M**atrix (**CEM**)

	v_1=0	v_2=1	v_3=10	v_4=4
u_1=2	.	.	-1	1
u_2=-3	4	2	-1	.
u_3=5	.	2	.	.

Step v: If any of the cell evaluation matrix values are negative the basic feasible solution is not optimal.

In this example, since two CEM values are negative, steps are to be taken to obtain an optimal solution.

Step IV: Iterate towards an Optimal Solution.

Step i: From the CEM, identify the cell with the most -ve entry. Here two cells have same -ve entry (-1), we choose the cell (1,3)

Step ii: Write down the initial feasible solution. Tick mark (√)the empty cell for which the cell evaluation is most -ve. This cell is known as **Identified cell**.

1	5	√	
			1
6		3	1

Step iii: Trace a path in this matrix consisting of series of alternately horizontal & vertical lines. This path begins & terminates in the identified cell. All the corners of the path lie in the cells for which allocations have been made. The path may skip over any number of occupied or vacant cells.

1	5	√	
			1
6		3	1

Step iv: Mark the identified cell as +ve & each occupied cell at the corners of the path alternately -ve, +ve, -ve and so on.

-1	5	√+	
			1
6+		-3	1

Step v: Make new allocation in the identified cell by entering the smallest allocations on the path that has been assigned a -ve sign. Add & subtract this new allocation from the cells of the path maintaining the row & column requirements. This causes one basic cell to become zero & other cells remain non-negative. The basic cell whose allocations has been made zero, leaves the solution.

1 - 1	5	1	
			1
6+1		3 - 1	1

→

	5	1	
			1
7		2	1

2nd Feasible Solution

The total cost for 2nd feasible solution is Rs. 101/-

Step V: Check for optimality

This can be done by repeating the steps done already.

a. m+n-1 = no of allocations
   3+4-1 = 6
   6 = 6 √

b. These (m+n-1) allocations are in independent positions

Step i: Set up a cost matrix containing the cost associated with the cells for which allocations have been made.

Step ii: Enter a set of number v_j along the top of the matrix & set of number u_i along left side of matrix so that their sum is equal to costs entered in the matrix of step i, shown below.

	v_1=0	v_2=2	v_3=10	v_4=4

u_1=1		3	11	
u_2=-3				1
u_3=5	5		15	9

Step iii: Fill the vacant cells with the sums of u_i & v_j.

	v_1=0	v_2=2	v_3=10	v_4=4
u_1=1	1	.		5
u_2=-3	-3	-1	7	.
u_3=5	.	7	.	.

Step iv: Subtract the cells values of this from the original cost matrix.

1	.	.	2
4	1	-1	.
.	1	.	.

Step v: Since one cell value is – ve, the 2nd feasible solution is not optima.

Step VI: Iterate towards an optimal solution

Step i: Identify the cell with the most -ve entry & the cell is (2,3).

Step ii: Write down again feasible solution. Mark the empty cell for which evaluation is -ve. This is called identified cell.

Step iii: trace the path.

Step iv: Mark +ve & -ve alternately.

	5	1	
		+√	-1
7		-2	+1

Step v: Make the new allocations in the identified cell by entering the smallest allocations in the identified cell by entering the smallest allocation on the path which has been assigned -ve sign. Subtract & add this amount from the cells.

	5	1	
		+1	1-1
7		2-1	1+1

→

	5	1	
		1	
7		1	2

3rd Feasible Solution(CEM)

The 3rd feasible solution, total transportation cost is Rs. 100/-

Step VII: Test for optimality

For above feasible solution

a. Number of allocations is m+n-1 i.e 6
b. These m+n-1 allocations are in independent positions

Step i: Set up the cost matrix containing +ve associated with the cells for allocations have been made.

Step ii: Enter a set of no. u_i & v_j.

	v_1=0	v_2=2	v_3=10	v_4=4
u_1=1		3	11	
u_2=-4			6	
u_3=5	5		15	9

Step iii: Fill the vacant cells.

	v_1=0	v_2=2	v_3=10	v_4=4

u_1=1	1	.	.	5
u_2=-3	-4	-2	.	0
u_3=5	.	7	.	.

Step iv: Subtract the cells values of the above matrix from the original cost matrix.

2+1	.	.	7-5
1+4	0+2	.	1-0
.	8-7	.	.

→

3	.	.	2
5	2	.	1
.	1	.	.

Cell Evaluation Matrix

Step v: Since all the cell values of CEM are +ve, the 3rd feasible solution is the optimal solution.

Total transportation Cost is = Rs. 100/-

Unbalanced Supply and Demand:

Problem:

		Warehouse			Supply
		W_1	W_2	W_3	
Plant	A	28	17	26	500
	B	19	12	16	300
Demand		250	250	500	

Solution:

The total demand is 1000, whereas the total supply is 800.

Σ Si (for i=1 to 2) < Σ Dj (for j=1 to 3)

To solve the problem, introduce an additional row with transportation cost zero indicating the unsatisfied demand.

		Warehouse			Supply
		W_1	W_2	W_3	
Plant	A	28	17	26	500
	B	19	12	16	300
Unsatisfied Demand		0	0	0	200
Demand		250	250	500	

Using least cost entry method, we get the following allocations.

		Warehouse			Supply
		W_1	W_2	W_3	
Plant	A	28(50)	17	26(450)	~~500~~
	B	19	12(250)	16(50)	~~300~~
Unsatisfied Demand		0(200)	0	0	~~200~~ 0
Demand		~~250~~	~~250~~	~~500~~	

Initial basic feasible solution

50 X 28 + 450 X 26 + 250 X 12 + 50 X 16 + 200 X 0 = 16900.

The remaining procedure remains as it is for finding optimal solution.

1. A product is manufactured at four factories A, B, C and D. Their unit production costs are Rs 2, Rs 3, Re 1 and Rs 5, respectively. Their production capacities are 50, 70, 30 and 50 units, respectively. These factories supply the product to four stores, demands of which are

25, 35, 105 and 20 units respectively. Unit transportation cost in rupees from each factory to each store is given in the table below.

		Stores			
		I	*II*	*III*	*IV*
Factories	*A*	2	4	6	11
	B	10	8	7	5
	C	13	3	9	12
	D	4	6	8	3

Determine the extent of deliveries from each of the factories to each of the stores, so that the total production and transportation cost is the minimum.

Solution:

The new transportation costs that include both the production and the transportation costs is given in Table

	I	*II*	*III*	*IV*	*Supply*
A	2 + 2 = 4	4 + 2 = 6	6 + 2 = 8	11 + 2 = 13	50
B	10 + 3 = 13	8 + 3 = 11	7 + 3 = 10	5 + 3 = 8	70
C	13 + 1 = 14	3 + 1 = 4	9 + 1 = 10	12 + 1 = 13	30
D	4 + 5 = 9	6 + 5 = 11	8 + 5 = 13	3 + 5 = 8	50
Demand	25	35	105	20	200 / 185

Since the total supply of 200 units exceeds the total demand of 185 units by 15 units, a dummy destination (store) is added (or created) to absorb the excess capacity. The associated cost coefficients in dummy store are taken as zero. This may be due to the reason that the surplus quantity remains lying in the respective factories and is not shipped at all. The modified table is shown in Table

	I	*II*	*III*	*IV*	*Dummy*	*Supply*
A	4 (25)	6 (5)	8 (20)	13	0	50
B	13	11	10 (70)	8	0	70
C	14	4 (30)	10	13	0	30
D	9	11	13 (15)	8 (20)	0 (15)	50
Demand	25	35	105	20	15	200

Using the VAM method the initial solution is shown in Table 9.29. It can be seen that 15 units are allocated to dummy store from factory D. This means that the company may cut down the production by 15 units at the factory that is proving to be uneconomical. Now to test the optimality of the solution shown in Table, evaluate each unoccupied cell in terms of opportunity cost associated with it as shown in Table

	I	*II*	*III*	*IV*	*Dummy*	*Supply*	u_i
A	4 (25)	6 (5)	8 (20)	13 +10	0 +5	50	$u_1 = -5$
B	13 +7	11 +3	10 (70)	8 +3	0 +3	70	$u_2 = -3$
C	14 +12	4 (30)	10 +4	13 +12	0 +7	30	$u_3 = -7$
D	9 0	11 0	13 (15)	8 (20)	0 (15)	50	$u_4 = 0$
Demand	25	35	105	20	15	200	
v_j	$v_1 = 9$	$v_2 = 11$	$v_3 = 13$	$v_4 = 8$	$v_5 = 0$		

Since the opportunity cost in all the unoccupied cells is positive, the initial solution is an optimal solution. The total cost of transportation associated with this solution is

Total cost = 4 × 25 + 6 × 5 + 8 × 20 + 10 × 70 + 4 × 30 + 13 × l5 + 8 × 20 + 0 × l5

= Rs l,465

Degeneracy: If the basic feasible solution of a transportation problem with m origins and n destinations has fewer than m + n – 1 positive x_{ij} (occupied cells), the problem is said to be a degenerate transportation problem.

Practice Problem:

A company has four warehouses and six stores. The wear houses altogether have a surplus of 22 units of a given commodity, divided among them as follows:

Wearhouses	: 1	2	3	4
Surplus	: 5	6	2	9

The six stores need 22 units of commodity. Individual requirements at these stores are 4, 4, 6, 2, 4 & 2 units respectively. Cost of shipping one unit of commodity from warehouse i to store j in rupees is given the matrix below.

		Stores						Supply
		1	2	3	4	5	6	
Warehouse	1	9	12	9	6	9	10	5
	2	7	3	7	7	5	5	6
	3	6	5	9	11	3	11	2
	4	6	8	11	2	2	10	9
Demand		4	4	6	2	4	2	

How should the products be shipped from the warehouse to store so that the transportation cost is minimum.

Solution:

Step I: Find initial basic feasible solution by VAM

		Stores						Supply
		1	2	3	4	5	6	
Warehouse	1	9	12	9(5)	6	9	10	5
	2	7	3(4)	7	7	5	5(2)	6
	3	6(1)	5	9(1)	11	3	11	2
	4	6(3)	8	11	2(2)	2(4)	10	9
Demand		4	4	6	2	4	2	22

Cost of transportation Z = Rs. 112/-

Step II: Optimality Test

a. m+n-1 = no. of allocations
   4+6-1 = 8
   / 9 = 8

Optimality test cannot be performed. Such solution is called degenerate solution.

b. These m+n-1 allocations must be in independent positions.

Now allocate an infinitesimally small but positive value ε to least cost of the cell. Subscripts are used when more than one such letter is required. Out of the unoccupied cell. Cell (3,5) has the least cost of Rs. 3. The ε should be made in this cell.

However, allocating ε to this cell forms a closed loop among cells (3,1), (3,5), (4,5) & (4,1) so that allocation in these cells do not remain in independent positions. Therefore, no allocation is made in cell (3,5).

There are two next higher cost cells with each cost of Rs. 5. Allocations in either of these cells does not result in closed loop. Hence ε can be made in either of these two cells. Let us make the allocation in cell (2,5).

	1	**2**	**3**	**4**	**5**	**6**
1			5			
2		4			ε	2
3	1		1			
4	3			2	4	

Now optimality can be applied.

	1	**2**	**3**	**4**	**5**	**6**
1			9			
2		3			5	5
3	6		9			
4	6			2	2	

u_i and v_j values:

		v_1=0	v_2=-6	v_3=3	v_4=-4	v_5=-4	v_6=-4
		1	**2**	**3**	**4**	**5**	**6**
u_1=6	**1**			9			
u_2=9	**2**		3			5	5
u_3=6	**3**	6		9			
u_4=6	**4**	6			2	2	

$u_i + v_j$ values in empty cells:

		v_1=0	v_2=-6	v_3=3	v_4=-4	v_5=-4	v_6=-4
		1	**2**	**3**	**4**	**5**	**6**
u_1=6	**1**	6	0	.	2	2	2
u_2=9	**2**	9	.	12	5	.	.
u_3=6	**3**	.	0	.	2	2	2
u_4=6	**4**	.	0	9	.	.	2

$c_{ij} - (u_i + v_j)$: **Cell Evaluation Matrix**

		v_1=0	v_2=-6	v_3=3	v_4=-4	v_5=-4	v_6=-4
		1	**2**	**3**	**4**	**5**	**6**
u_1=6	**1**	3	12	.	4	7	8
u_2=9	**2**	-2	.	-5	2	.	.
u_3=6	**3**	.	5	.	9	1	9
u_4=6	**4**	.	8	2	.	.	8

Since cells (2,1) & (2,3) have - ve values, the current feasible solution is not optimal.

Step III: Iterate towards an optimal solution

	1	2	3	4	5	6
1			5			
2		4	√ +		ε -	2
3	1+		1-			
4	3-			2	4 +	

	1	2	3	4	5	6
1			5			
2		4	+ε		ε-ε	2
3	1+ε		1-ε			
4	3-ε			2	4+ε	

2nd Feasible solution:

	1	2	3	4	5	6
1			5			
2		4	ε			2
3	1		1			
4	3			2	4	

Cost of transportation Z = Rs. 112/-

MODI method:

		v_1=0	v_2=-1	v_3=3	v_4=-4	v_5=-4	v_6=1
		1	**2**	**3**	**4**	**5**	**6**
u_1=6	**1**			9			
u_2=4	**2**		3	7			5
u_3=6	**3**	6		9			
u_4=6	**4**	6			2	2	

c_{ij} - (u_i + v_j): **Cell Evaluation Matrix**

	1	2	3	4	5	6
1	3	7	.	4	7	3
2	3	.	.	7	5	.
3	.	0	.	9	1	4
4	.	3	2	.	.	3

Since all the cell values are + ve, the 2nd feasible solution is an optimal solution. Hence total transportation cost is Rs. 112/-.

Degeneracy at subsequent iterations: To resolve degeneracy, which occurs during optimality test, the quantity may be allocated to one or more cells that have recently been unoccupied, to have m + n – 1 number of occupied cells in the new solution.

Goods have to be transported from sources S1, S2 and S3 to destinations D1, D2 and D3. The transportation cost per unit, capacities of the sources, and the requirements of the destinations are given in the following table.

	D_1	D_2	D_3	*Supply*
S_1	8	5	6	120
S_2	15	10	12	80
S_3	3	9	10	80
Demand	150	80	50	

Determine a transportation schedule so that cost is minimized.

Solution Using North-West Corner Method, the non-degenerate initial basic feasible solution so obtained is given in Table

	D_1	D_2	D_3	Supply
S_1	8 (120)	5	6	120
S_2	15 (30)	10 (50)	12	80
S_3	3	9 (30)	10 (50)	80
Demand	150	80	50	280

	v1 = 0	v2 = -5	v3 = -4
u1= 8	8	3	4
u2= 15	15	10	11
u3= 14	14	9	10

	v1 = 0	v2 = -5	v3 = -4
u1= 8		2	2
u2= 15			1
u3= 14	-11		

Cell Evaluation Matrix

Since the values of CEM are negative, the IBFS is not optimal
Iteration towards optimality test:

120		
30-	+50	
√+	-30	50

120		
ε	80	
30		50

2nd Feasible Solution

Total Transportation Cost = Rs. 2350/-
Optimality Check:

m + n - 1 = no of allocations
3 + 3 - 1 = 4
5 ≠ 4

After allocating the ε all the values are in independent position

	v1 = 0	v2 = -5	v3 = 7
u1= 8	8	3	15
u2= 15	15	10	22
u3= 3	3	-2	10

	v1 = 0	v2 = -5	v3 = 7
u1= 8		2	-9
u2= 15			-10
u3= 3		11	

CEM

Since the values of CEM are negative, the 2nd feasible solution is not optimal

120		

-ε	80	+√
+30		-50

120		
	80	
30		50

To test the optimality of the solution given in Table, calculate ui, vj and dij as usual as shown in Table

	D_1	D_2	D_3	Supply	u_i
S_1	8 (120)	5 +2	6 +2	120	$u_1 = -7$
S_2	15 (–) (30)	10 (50) (+)	12 +1	80	$u_1 = 0$
S_3	3 (+) −11	9 (30) (–)	10 (50)	80	$u_1 = -1$
Demand	150	80	50	280	
v_j	$v_1 = 15$	$v_2 = 10$	$v_3 = 11$		

Since the unoccupied cell (S_3, D_1) has the largest negative opportunity cost of – 11, therefore, cell (S_3, D_1) is entered into the new solution mix. The closed path for (S_3, D_1) is shown in above Table.

The maximum allocation to (S_3, D_1) is 30. However, when this amount is allocated to (S_3, D_1) both cells (S_2, D_1) and (S_3, D_2) become unoccupied because these two have same allocations. Thus, the number of positive allocations become less than the required number, m + n – 1 = 3 + 3 – 1 = 5. Hence, this is a degenerate solution as shown in Table

	D_1	D_2	D_3	Supply
S_1	8 (120)	5	6	120
S_2	15	10 (80)	12	80
S_3	3 (30)	9 (Δ)	10 (50)	80
Demand	150	80	50	280

To remove degeneracy a quantity Δ is assigned to one of the cells that has become unoccupied so that there are m + n – 1 occupied cells. Assign Δ to (S_3, $_{D2)}$ and proceed with the usual solution procedure. The optimal solution is given in Table

	D_1	D_2	D_3	Supply
S_1	8 (70)	5 (Δ)	6 (50)	120
S_2	15	10 (80)	12	80
S_3	3 (80)	9	10	80
Demand	150	80	50	280

The total minimum transportation cost associated with this solution is:

Total cost = 8 × 70 + 6 × 50 + 10 × 80 + 3 × 80

= Rs 1,900

120	ε	

	80	
30		50

8	5	
	10	
3		10

	v1 = 0	v2 = -3	v3 = 7
u1= 8	8	5	15
u2= 13	13	10	20
u3= 3	3	0	10

		-9
2		-8
	9	

2		1
	9	11

TTC = Rs. 1900/-

Unbalanced Supply and Demand:

Practice Problems:

1. A manufacturer wants to ship 22 loads of his product as shown below. The matrix gives the kilometers from sources of supply to the destinations.

	D_1	D_2	D_3	D_4	D_5	supply
S_1	5	8	6	6	3	8
S_2	4	7	7	6	5	5
S_3	8	4	6	6	4	9
Demand	4	4	5	4	8	

Shipping costs are Rs 10 per load per km. What shipping schedule should be used to minimize total transportation cost?

	D_1	D_2	D_3	D_4	D_5	supply					
S_1	~~5~~	~~8~~	~~6~~	~~6~~	~~3(8)~~	8 0	2	2	2		
S_2	~~4(4)~~	~~7~~	7	6(1)	~~5~~	5 1 0	1	1	1	2	1
S_3	~~8~~	~~4(4)~~	6(2)	6(3)	~~4~~	9 5 3 0	0	0	2	2	0
Dummy	~~0~~	~~0~~	~~0(3)~~	~~0~~	~~0~~	3 0	0				
Demand	4 0	4 0	5 2 0	4 1 0	8 0						
	4	4	6	6	4						
	1	3	0	0	1						
	1		0	0	1						
	4		1	0							
			1	0							

TTC = Rs. 3x8+4x4+6x1+4x4+6x2+6x3+0x3
=Rs. 24+16+6+16+12+18
=Rs. 92

4+5-1=7

$8 \neq 7$

	D_1	D_2	D_3	D_4	D_5
S_1	~~5~~	~~8~~	~~6~~	~~6~~	~~3(8)~~
S_2	~~4(4)~~	~~7~~	7	6(1)	~~5~~
S_3	~~8~~	~~4(4)~~	6(2)	6(3)	~~4~~
	~~0~~	~~0~~	~~0(3)~~	~~0~~	~~0~~

	0	0	2	2	2
1	1	1	3	3	3
4	4	4	6	6	6
4	4	4	6	6	6
-2	-2	-2	0	0	0

	0	0	2	2	2
1	4	7	3	3	.
4	.	3	1	.	-1
4	4	.	.	.	-2
-2	2	2	.	0	.

				8
4			1	
	4	2	~~3~~	√
		3		~~ε~~

				8
4			1	
	4	2	3	
		3		0

2nd Feasible Solution

TTC = Rs. 92/-

Repeat the steps upto CEM

4	7	3	3	
	3	1		1
4				-2
2	2		0	

MAXIMIZATION TRANSPORTATION PROBLEM

A company has four manufacturing plants and five warehouses. Each plant manufactures the same product, which is sold at different prices in each warehouse area. The cost of manufacturing and cost of raw materials are different in each plant due to various factors. The capacities of the plants are also different. The relevant data is given in the following table:

Item	Plant 1	2	3	4
Manufacturing cost (Rs) per unit	12	10	8	8
Raw material cost (Rs) per unit	8	7	7	5
Capacity per unit time	100	200	120	80

The company has five warehouses. The sale prices, transportation costs and demands are given in the following table:

Warehouse	Transportation Cost (Rs) per Unit 1	2	3	4	Sale Price	Demand per Unit (Rs)
A	4	7	4	3	30	80
B	8	9	7	8	32	120
C	2	7	6	10	28	150
D	10	7	5	8	34	70
E	2	5	8	9	30	90

(a) Formulate this problem as a transportation problem in order to maximize profit.
(b) Find the solution using VAM method.
(c) Test for optimality and find the optimal solution.

Solution Based on the given data, the profit matrix can be derived by using following equation.

Profit = Sales price – Manufacturing cost – Raw material cost – Transportation cost
Profit (A,1) = 30 - 12 - 8 - 4 = 6

Profit Matrix:

	1	2	3	4
A	6	6	11	14
B	4	6	10	11
C	6	4	7	5
D	4	10	14	13
E	8	8	7	8

Loss Matrix:

	1	2	3	4
A	-6	-6	-11	-14
B	-4	-6	-10	-11
C	-6	-4	-7	-5
D	-4	-10	-14	-13
E	-8	-8	-7	-8

Choose the highest value among the Profit matrix and subtract all other values from 14

	1	2	3	4	
A	8	8	3	0	
B	10	8	4	3	
C	8	10	7	9	
D	10	4	0	1	
E	6	6	7	6	

	1	2	3	4	Dummy	Demand
A	6	6	11	15	0	80
B	4	6	10	12	0	120
C	6	4	7	6	0	150
D	4	10	14	14	0	70
E	8	8	7	9	0	90
Supply	100	200	120	80	10	510

Converting data on profit into cost by subtracting all the profit values in the table from the highest profit value. Since highest profit value is 15, therefore, subtracting all cell values including itself from it. The problem now becomes a usual cost minimizing transportation problem.

	1	2	3	4	Dummy	Demand	u_i
A	9 +7	9 +5	4 +4	0 (80)	15 +12	80	$u_1 = 4$
B	11 +4	9 (70)	5 (–) (50)	3 (+) −2	15 +7	120	$u_2 = 9$
C	9 (100)	11 (40)	8 +1	11 +4	15 (10)	150	$u_3 = 11$
D	11 +8	5 0	1 (+) (70)	1 (Δ) (–)	15 +11	70	$u_4 = 5$
E	7 +2	7 (90)	8 +5	6 +3	15 +9	90	$u_5 = 7$
Supply	100	200	120	80	10	510	
v_j	$v_1 = -2$	$v_2 = 0$	$v_3 = -4$	$v_4 = -4$	$v_5 = -1$		

Since at initial solution stage, the number of occupied cells are only 8, which is one less than the required number, m+n– 1 = 9, therefore, the solution is degenerate.

However, after making an allocation of Δ to the cell (D, 4), the initial solution has now become eligible for optimality test.

Apply MODI method to evaluate each unoccupied cell in terms of opportunity cost associated with it in the usual manner.

The cell (B, 4) has a negative opportunity cost (i.e. – 2) as shown in Table.

Introduce it into the new solution by constructing a loop shown in Table 9.52. The new solution is given in Table 9.53, where Δ has shifted from cell (D, 4) to cell (B, 4).

	1	2	3	4	Dummy	Demand	u_i
A	9 +5	9 +3	4 +2	0 (80)	15 +5	80	$u_1 = -5$
B	11 +4	9 (70)	5 (50)	3 (Δ)	15 +2	120	$u_2 = -2$
C	9 (100)	11 (40)	8 +1	11 +6	15 (10)	150	$u_3 = 0$
D	11 +8	5 0	1 (70)	1 +2	15 +6	70	$u_4 = -6$
E	7 +2	7 (90)	8 +5	6 +5	15 +4	90	$u_5 = -4$
Supply	100	200	120	80	10	510	
v_j	$v_1 = 9$	$v_2 = 11$	$v_3 = 7$	$v_4 = 5$	$v_5 = 15$		

Since there is no negative opportunity cost in any of the unoccupied cells in Table, therefore, this solution is the optimal solution. However, the zero opportunity cost in cell (D, 2) indicates the existence of an alternative solution. The total maximization profit associated with the solution is

Total profit = 9 × 70 + 5 × 50 + 9 × 100 + 11 × 40 + 15 × 10 + 1 × 70 + 7 × 90

= Rs 4,580

ASSIGNMENT PROBLEMS

The assignment problem is a special type of transportation problem, where the objective is to minimize the cost or time of completing a number of jobs by a number of persons. In other words, when the problem involves the allocation of n different facilities to n different tasks, it is often termed as an assignment problem. The model's primary usefulness is for planning. The assignment problem also encompasses an important sub-class of so-called shortest- (or longest-) route models. The assignment model is useful in solving problems such as, assignment of machines to jobs, assignment of salesmen to sales territories, travelling salesman problem, etc.

It may be noted that with n facilities and n jobs, there are n! possible assignments. One way of finding an optimal assignment is to write all the n! possible arrangements, evaluate their total cost, and select the assignment with minimum cost. But, due to heavy computational burden this method is not suitable. This chapter concentrates on an efficient method for solving assignment problems that was developed by a Hungarian mathematician D.Konig.

Formulation of an Assignment Problem:

To formulate the assignment problem in mathematical programming terms, we define the activity variables as

$$x_{ij} = \begin{cases} 1 & \text{if job } j \text{ is performed by worker } i \\ 0 & \text{otherwise} \end{cases}$$

for i = 1, 2, ..., n and j = 1, 2, ..., n

In the above table, c_{ij} is the cost of performing jth job by ith worker.

The optimization model is

Minimize $c_{11}x_{11} + c_{12}x_{12} + \text{--------} + c_{nn}x_{nn}$

Subject to

$x_{i1} + x_{i2} + \text{.....} + x_{in} = 1 \quad _i = 1, 2, \text{.....}, n$

$x_{1j} + x_{2j} + \text{.....} + x_{nj} = 1 \quad _j = 1, 2, \text{.....}, n$

$x_{ij} = 0 \text{ or } 1$

Assumptions:

i. Number of jobs is equal to the number of machines or persons.
ii. Each man or machine is assigned only one job.
iii. Each man or machine is independently capable of handling any job to be done.
iv. Assigning criteria is clearly specified (minimizing cost or maximizing profit).

Hungarian Method:

It is an efficient method for solving assignment problems. This method is based on the following principle:

"If a constant is added to, or subtracted from, every element of a row and/or a column of the given cost matrix of an assignment problem, the resulting assignment problem has the same optimal solution as the original problem".

Algorithm: The objective of this section is to examine a computational method – an algorithm – for deriving solutions to the assignment problems.

Steps:

i. Identify the minimum element in each row and subtract it from every element of that row.
ii. Identify the minimum element in each column and subtract it from every element of that column.
iii. Make the assignments for the reduced matrix obtained from steps 1 and 2 in the following way:
    a. For each row or column with a single zero value cell that has not be assigned or eliminated, box that zero value as an assigned cell.
    b. For every zero that becomes assigned, cross out (X) all other zeros in the same row and the same column.
    c. If for a row and a column, there are two or more zeros and one cannot be chosen by inspection, then you are at liberty to choose the cell arbitrarily for assignment.
    d. The above process may be continued until every zero cell is either assigned or crossed (X).
iv. An optimal assignment is found, if the number of assigned cells equals the number of rows (and columns). In case you have chosen a zero cell arbitrarily, there may be alternate optimal solutions. If no optimal solution is found, go to step v.
v. Draw the minimum number of vertical and horizontal lines necessary to cover all the zeros in the reduced matrix obtained from step 3 by adopting the following procedure:
    a. Mark all the rows that do not have assignments.
    b. Mark all the columns (not already marked) which have zeros in the marked rows.
    c. Mark all the rows (not already marked) that have assignments in marked columns.
    d. Repeat steps 5 (a) to (c) until no more rows or columns can be marked.
    e. Draw straight lines through all unmarked rows and marked columns.
vi. Select the smallest element from all the uncovered elements. Subtract this smallest element from all the uncovered elements and add it to the elements, which lie at the intersection of two lines. Thus, we obtain another reduced matrix for fresh assignment.

Go to step iii and repeat the procedure until you arrive at an optimal assignment.

Maximization Case: There are problems where certain facilities have to be assigned to a number of jobs, so as to maximize the overall performance of the assignment. The Hungarian Method can also solve such problems, as it is easy to obtain an equivalent minimization problem by converting every number in the matrix to an opportunity loss. The conversion is accomplished by subtracting all the elements of the given matrix from the highest element. It turns out that minimizing opportunity loss produces the same assignment solution as the original maximization problem.

Unbalanced Assignment Problems: In the previous section, the number of persons and the number of jobs were assumed to be the same. In this section, we remove this assumption and consider a situation where the number of persons is not equal to the number of jobs. In all such cases, fictitious rows and/or columns are added in the matrix to make it a square matrix. Then, we apply the usual Hungarian Method to this resulting balanced assignment problem. We provide the following example to illustrate the solution of an unbalanced assignment problem.

Travelling Salesman Problem

TSP Hungarian Method

- **Step 1:**
  - In the cost (or distance) matrix, find the minimum cost in each row. Subtract each cost by the minimum cost in its row
  - In the new matrix, find the minimum cost in each column. Subtract each cost by the minimum cost in its column
- **Step 2:**
  - Calculate the penalties of all zeros by adding the min value (other than self) in the same row and the min value (other than self) in the same col
  - Pick the zero with the largest penalty. Include arc from the row to the col in the tour
  - Cross out the corresponding row and col to formulate the remaining matrix
  - Stop if there is no row or col left; otherwise, go back to Step 1

SEORL SUBSCRIBE

1

	A	B	C	D	E	min
A	--	132	217	164	58	58
B	132	--	290	201	79	79
C	217	290	--	113	303	113
D	164	201	113	--	196	113
E	58	79	303	196	--	58

2

	A	B	C	D	E
A	--	74	159	106	0
B	53	--	211	122	0
C	104	177	--	0	190
D	51	88	0	--	83
E	0	21	245	138	--

3

	A	B	C	D	E
A	--	74	159	106	0
B	53	--	211	122	0
C	104	177	--	0	190
D	51	88	0	--	83
E	0	21	245	138	--
min	0	21	0	0	0

4

	A	B	C	D	E
A	--	53	159	106	0
B	53	--	211	122	0
C	104	156	--	0	190
D	51	67	0	--	83
E	0	0	245	138	--

7

	A	B	C	D	E
A	--	53	0		0
B	53	--	52		0
C				C→D	
D	0	16	--		32
E	0	0	86		--

8

	A	B	C	D	E
A	~~--~~	53	0(52)		0(0)
B	53	--	52		0(52)
C				C→D	
D	0(16)	16	--		32
E	0(0)	0(16)	86		--

A→C
C→A prohibited

9

	A	B	C	D	E
A			A→C		
B	53				~~0(85)~~
C				C→D	
D	0(16)	16			32
E	0(0)	0(16)			--

B→E
E→B prohibited

10

	A	B	C	D	E
A			A→C		
B					B→E
C				C→D	
D	0(0)	0(0)			
E	~~0(0)~~	--			

E→A
A→E prohibited

5

	A	B	C	D	E
A	--	53	159	106	0(53)
B	53	--	211	122	0(53)
C	104	156	--	0(210)	190
D	51	67	0(210)	--	83
E	0(51)	0(53)	245	138	--

11

	A	B	C	D	E
A			A→C		
B					B→E
C				C→D	
D		0			
E	E→A				

D→B
B→D prohibited

6

	A	B	C	D	E
A	--	53	159	106	0(53)
B	53	--	211	122	0(53)
C	~~104~~	~~156~~	--	0(210)	~~190~~
D	51	67	0(210)	--	83
E	0(51)	0(53)	245	138	--

C→D
D→C prohibited

12

	A	B	C	D	E
A	--	132	217	164	58
B	132	--	290	201	79
C	217	290	--	113	303
D	164	201	113	--	196
E	58	79	303	196	--

A→C→D→B→E→A
Min distance: 217+113+201+79+58=668